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Comparing Portfolio Performance to a Benchmark: Risk and Attribution Analyses

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Abstract

In recent times, the risk-adjusted return performance of a fund or a portfolio compared to that of a benchmark or reference index is a concern for the investors and their brokers. The difference between these two risk-adjusted returns is known as “risk adjusted excess return or differential return”. In addition to risk analysis, the sources of or reasons for the differential return calls for a systematic “differential attribution analysis”. To compare risk-adjusted return of a benchmark with that of a portfolio return (that is, differential return), risk adjustment is done using Sharpe’s standard deviation, Treynor’s beta, or Jensen’s alpha. To measure the differential return, it becomes obvious whether the portfolio is riskier than the benchmark or not. Attribution analysis measures the difference between the risk-adjusted portfolio return and that of the benchmark returns into differences due to allocating the portfolio sector weights differently than the benchmark (the difference can either be positive or be negative) and then identifies the reasons or sources of such difference. That is, attribution analysis analyzes the reasons for over or under performance of a portfolio in terms of its risk-adjusted return compared to that of a benchmark. The two-stage attribution analysis uses either Jensen’s alpha or Fama’s beta to adjust for taking on unsystematic risk. This article proposes a risk adjusted performance attribution analysis that incorporates risk measures with the Brinson model (1986) that is used to perform performance attribution by decomposing the differential return of a portfolio ignoring risk. Traditional attribution analysis could conclude that a manager has done well due to taking on more risk or has underperformed in a less risky investment.

I. Introduction

Attribution analysis is an evaluation tool used to explain and analyze a portfolio’s (or portfolio manager’s) performance, especially against a particular benchmark. Attribution analysis focuses on three factors: the manager’s investment picks and asset allocation, their investment style, and the market timing of their decisions and trades. Attribution analysis is a sophisticated method for evaluating the performance of a

portfolio or fund manager. Also known as “return attribution” or “performance attribution,” it attempts to quantitatively analyze the efficiency of an active fund manager in investment selections and decisions—and to identify the resulting performance of a portfolio in terms of risk-adjusted excess returns, especially as compared to an index or another benchmark (Investopedia, 2021).

According to Investopedia.com. (2021) For portfolio managers and investment firms, attribution analysis can be an effective tool to assess strategies. For investors, attribution analysis works to assess the performance of fund or money managers.

- Attribution analysis is an evaluation tool used to explain and analyze a portfolio's (or portfolio manager's) performance, especially against a particular benchmark.
- Attribution analysis focuses on three factors: the manager's investment picks and asset allocation, their investment style, and the market timing of their decisions and trades.
- Asset class and weighting of assets within a portfolio figure in analysis of the investment choices.
- Investment style reflects the nature of the holdings: low-risk, growth-oriented, etc.
- The impact of market timing is hard to quantify, and many analysts rate it as less important in attribution analysis than asset selection and investment style.

In recent days, research on investment performance shows excess or differential return of a portfolio relative to a benchmark return, given their respective risk levels. A portfolio may have either a positive or a negative risk-adjusted excess return compared to risk-adjusted benchmark return. A portfolio with positive differential return is performing better than the benchmark, and a portfolio with negative excess return is performing poor compared to the benchmark. Thus, this risk-adjusted excess return of a portfolio guides the investors whether to invest or not in a particular portfolio.

However, the risk-taking or risk-aversion behavior of an investor determines, to a significant extent, whether to invest in a riskier portfolio for higher excess return or a low-risk portfolio with low or no excess return. Therefore, like risk-return trade-off, investors do face risk-excess return trade-off. However, this over or under performance of a portfolio is not presented on a nominal basis, but on a risk-adjustment basis. Thus, it does properly consider the risk of the portfolio relative to that of the benchmark. Now, assuming that the market is efficient in the sense that the market can generate a SML and the benchmark risk = market risk, $\beta_M = 1.0$, and if the portfolio risk, β_P , is greater than one, then the portfolio is riskier than the benchmark. Let $\beta_P = 1.4$. If $R_p = 12\%$ and $R_M = 8\%$, and RFR = 0. Then the nominal excess return of the portfolio is 4.0%.

It is worth mentioning here that Fisher and D'Alessandro (2018) made a point that the "general practice [that] implicitly assume that the benchmark has a similar style and risk as the portfolio. Performance attribution analysis that uses this approach can result in misleading conclusions because it may appear the portfolio manager has done a good job in selecting individual assets when the higher returns were due to taking on more risk. Conversely, it could appear the manager did a poor job selecting assets when they took less risk than the benchmark". Whether this logic is correct or not depends on many factors and once the word "style" is dropped, it may become confusing. If, in our example, $\beta_P = 1.4$, the excess nominal return of the portfolio is = 4% = $(\bar{R}_P - \bar{R}_b) = \alpha_N$. Now, this 4% excess return cannot be attributed to the portfolio manager who has, apparently, done an excellent job. Here, nominal excess return and risk-adjusted excess return are same. Therefore, this (when nominal excess return and risk-adjusted excess return are same) will help understand what is going on in the portfolio market as well as in the benchmark return. Therefore, it is necessary for attribution of risk-adjusted excess return (sometimes known as differential return) of a portfolio to assume that the benchmark risk-level = market risk-level = $\beta_b = \beta_M = 1.0$.

The Capital Asset Pricing Model (CAPM) was introduced by Treynor (1961), Sharpe (1964), Lintner (1965), and Mossin (1966), building on the earlier work of Harry Markowitz's diversification and modern portfolio theory. In 1968, Jensen's alpha was introduced to measure mutual fund managers' performance. In 1985 and 1986, the Brinson performance attribution models were introduced to evaluate managers' investment decision making skills. Ankrum (1992) attempted to combine both risk measures and performance attribution, however fell short of reconciling to Jensen's alpha. Subsequently there have been a variety of attempts by Kophamel (2003), Obeid (2005), Menchero (2006/2007), Bacon (2008) and Spaulding (2016); however, none of these attempts incorporate both market risk and non-systematic risk. Bacon's model does not properly reconcile to the additional market risk (risk of having beta greater than or less than 1) because his expected portfolio return calculation uses individual sector benchmark returns rather than the overall benchmark return. Spaulding (2016) proposed a simple and intuitive risk adjusted performance attribution model using a lesser-known risk adjusted return, Modigliani & Modigliani (M2). M2; however, prices risk based on the portfolio's Sharpe ratio rather than the market's Sharpe ratio as is required by the CAPM. To date, no one has successfully combined CAPM, risk and attribution into one robust model. This article introduces a Risk Adjusted Performance Attribution (RAPA) model that builds on the work of all the previous mentioned authors by incorporating beta to calculate risk adjusted returns for each sector and using Fama's (1972) concept of "net selectivity" to calculate the non-systematic risk portion for portfolios that are less well diversified than the benchmark. The appropriate risk adjusted returns are then used as inputs to calculate the selection (and interaction) components of attribution analysis.

Brinson attribution models are widely used to evaluate performance by decomposing a manager's over or under active return into three main categories; allocation, selection, and interaction. These categories add up to the manager's active return contribution for each sector and the weighted sum of each sector's contribution to active return will

add to the overall portfolio over (under) performance. Brinson attribution helps answer the questions; (1) How well did the manager decide which sectors to invest in? (2) How well did the manager select individual investments? (3) Did the manager play up his/her strengths or play down his/her weaknesses? (4) Did the manager overweight bad performing sectors or under-weight good performing sectors? To implement this in-depth analysis, only the weights and returns are needed at the sector levels for both the portfolio and the benchmark. To do multi-period analysis, this information is needed every measurement period, for example, either monthly or quarterly.

Most practitioners, if not all, use Brinson attribution models with nominal returns for the manager's portfolio. As noted above, the results can be very misleading. Spaulding (2016) astutely observes that risk-adjusted returns, rather than nominal returns, should be used, and proposed substituting Modigliani-Modigliani (MM) for the manager's portfolio return in the Brinson model. This approach adjusts for risk but is not based on the CAPM which provides a theoretical basis for how the return should vary with the degree of risk. It also does not lend itself to separating risk into components (systematic vs. unsystematic due to less diversification) as was suggested by Fama (1972) and knowing how much of the return should be associated with each component. As we will show later, M2 is more like Fama's measure of total risk to adjust returns although the two can differ significantly. Thus, we propose an approach that uses beta to calculate risk-adjusted returns and then use what Tofallis (2006) refers to as a "beta estimator" and Bacon (2008) refers to as "Fama's beta" to further adjust the returns for diversification to arrive at what Fama (1972) refers to as "net selectivity."

The CAPM model expresses the return for a portfolio (or any asset) as follows:

$$\text{CAPM Portfolio Expected Return} = \text{Risk Free Rate} + \text{Beta} \times (\text{Market Return} - \text{Risk Free Rate}) = E(R_P) = RFR + \beta_P \times (R_M - RFR)$$

In practice, a suitable benchmark is used as a proxy for the market returns, thus we have

$$\text{CAPM Portfolio Expected Return} = \text{Risk Free Rate} + \text{Beta} \times (\text{Benchmark Return} - \text{Risk Free Rate}) = E(R_P) = RFR + \beta_P \times (R_B - RFR)$$

The difference between the portfolio's actual return and the portfolio's expected return is Jensen's alpha (Portfolio Actual Return – Portfolio Expected Return). Jensen's alpha is the over or under performance relative to the CAPM expected portfolio return (benchmark adjusted for the portfolio's beta). Jensen's alpha is therefore alpha after removing all the beta market risk.

Let us calculate and decompose Jensen alpha, α_J :

For simplicity, let us assume that the risk-free rate (RFR) = 0.

$$E(R_M) = 8\%, E(R_P) = 12\%, \beta_P = 1.4, \beta_M = 1 = \beta_B.$$

On the other hand, nominal alpha is the excess return relative to the market (benchmark) return without taking risk into account. The nominal excess return is calculated as actual Return – benchmark Return. Alam (2021) expressed it as $(\bar{R}_P - \bar{R}_M) = \alpha_N$.

Alam (2021) also attributed the difference between nominal alpha and Jensen's alpha $(\alpha_N - \alpha_J) = \lambda_i$. In our example, $\alpha_N = 4\%$.

How did the portfolio earn nominal alpha of 4.0% (12.0% - 8.0%)? The portfolio took on more risk than the benchmark. Based on the portfolio's historical beta, the portfolio should have earned an 11.2% (8.0% x 1.4%). On a risk adjusted basis, the portfolio achieved Jensen's alpha, α_J , of 0.6%. (=12.0%-11.2%). Since Jensen's alpha

represents the portion of alpha earned after adjusting for beta risk, the remainder is the portion of alpha earned on a risk adjusted basis. In the example, the portion of alpha due to beta risk is $3.2\% = (4\% - 0.6\%)$.

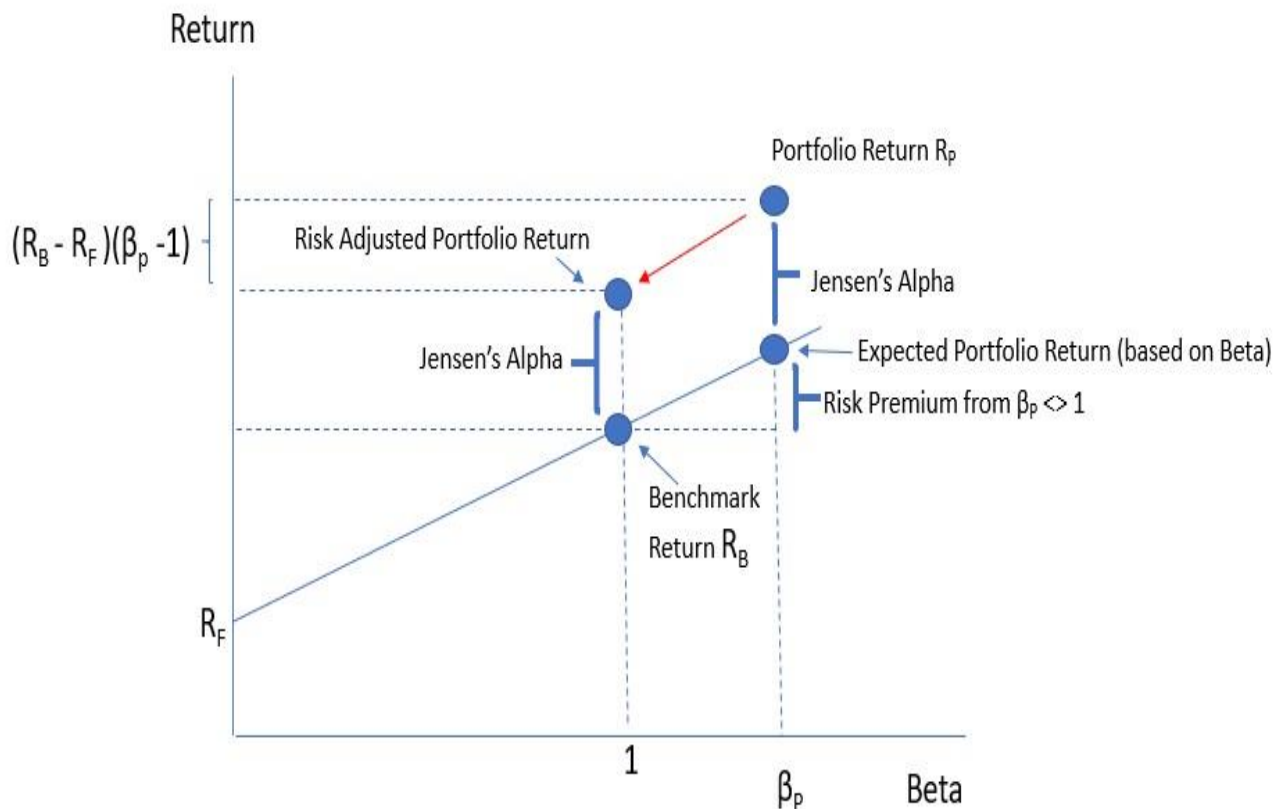
As we discussed earlier, for Brinson attribution we need to substitute the portfolio's nominal return with the portfolio's risk adjusted return. In this example, what is the portfolio's risk adjusted return? The 11.2% is the portfolio's expected return ($= 8\% \times 1.4$), but not its actual risk adjusted return. The portfolio's risk adjusted return can be calculated by removing the risk greater than or less than the market. The market or benchmark always has a beta of 1, therefore the portfolio risk adjusted return can be calculated by removing all beta risk from the portfolio that is greater than or less than 1. The portfolio's risk adjusted return is therefore $8.8\% (12.0\% - [8.0\% \times (1.4 - 1.0)])$. Once the portfolio risk adjusted return is determined, the risk adjusted alpha is simply the portfolio's risk adjusted return less the benchmark return. In this example, the 7.0% benchmark return is subtracted from the portfolio's risk adjusted return of 8.2%, resulting in a risk adjusted alpha of $3.2\% (= 8.8\% - 8.0\%)$. This 3.2% risk adjusted alpha (portfolio risk adjusted return less the benchmark return) is the same as Jensen's alpha calculated above (portfolio actual return minus portfolio expected return). This is the reconciliation of Jensen's alpha and Brinson attribution.

Exhibit I illustrates how the return is risk adjusted. The far-right vertical line that crosses the x-axis at the portfolio's beta represents the expected return formula. The top dot is the actual portfolio return (R_p). The bottom dot is the expected portfolio return which is determined by the benchmark return adjusted for the portfolio's beta. The difference between the actual return of the portfolio and the expected portfolio return is Jensen's alpha. The additional market risk premium resulting from the portfolio beta greater than or less than the benchmark beta of 1 is the difference between the nominal alpha and Jensen's alpha.

The left vertical line that crosses the x-axis at the beta of 1, represents the portfolio risk adjusted return less the benchmark return that is used in the Brinson model.

Exhibit I

Risk Adjusted Portfolio Return



Note that the difference between the risk adjusted portfolio return and the benchmark return is simply Jensen's alpha. Risk adjusting the return simply determines what the return would be if it had a beta of 1 but the portfolio still earned Jensen's alpha. It is the risk adjusted return that is then used in one of the Brinson attributions models which can be used to calculate the selection component (and interaction) for each

sector.¹ As we will see, the risk adjustment does not impact the allocation component of the attribution model.²

The two Brinson models used in the industry are BHB and BF, Brinson-Hood-Bee bower and Brinson-Fachler, respectively. BHB and BF are identical, except for the individual sector allocation scores. The following provides a summary of the math for each.

Sector Selection is calculated as the difference in sector returns multiplied by the benchmark sector weight $(\text{Portfolio Sector Return} - \text{Benchmark Sector Return}) * \text{Benchmark Sector Weight}$.

Interaction is calculated as the difference in sector returns multiplied by the difference in sector weights $(\text{Portfolio Sector Weight} - \text{Benchmark Sector Weight}) * (\text{Portfolio Sector Return} - \text{Benchmark Sector Return})$.

BHB Sector Allocation is calculated as the difference in sector weights multiplied by the benchmark sector return $(\text{Portfolio Sector Weight} - \text{Benchmark Sector Weight}) * \text{Benchmark Sector Return}$.

BF Sector Allocation is calculated as the difference in sector weights multiplied by the difference in the benchmark sector return less the overall benchmark return $(\text{Portfolio Sector Weight} - \text{Benchmark Sector Weight}) * (\text{Benchmark Sector Return} - \text{Overall Benchmark Return})$.

With BHB, positive sector allocation scores are achieved when the portfolio over-weights a positive performing benchmark sector or under-weights a negative performing benchmark sector. With BF, positive sector allocation scores are achieved when the portfolio over-weights a benchmark sector that over performs the overall

¹ The Ankrim model cited earlier erred by using the expected return from the CAPM as the input for the selection component in the attribution analysis. This is not the same as the risk-adjusted return and is the reason that his model does not reconcile to alpha. Ankrim's model removes some of the alpha and understates what the manager has earned from selection.

² Bacon adjusts some of the beta risk in his model out of allocation. We believe it should be part of net selection that is consistent with Fama (1972).

benchmark, or under-weights a benchmark sector that under performs the overall benchmark return. The total allocation score for both Brinson models will always be the same, only the sector allocation scores will vary.

Since Brinson-Fachler is generally used by most practitioners, an example of Brinson-Fachler attribution follows. Exhibit II is the typical data set used to perform the analysis.

Exhibit II

A	B	C	D	E	F	G
2		Portfolio Weights	Benchmark Weights	Portfolio Returns	Benchmark Returns	Nominal Alpha
3	Sector A	53.0%	23.0%	12.0%	9.0%	
4	Sector B	8.0%	2.0%	5.0%	8.0%	
5	Sector C	9.0%	14.0%	16.0%	13.0%	
6	Sector D	14.0%	36.0%	9.0%	10.0%	
7	Sector E	16.0%	25.0%	14.0%	12.0%	
8	Total	100.0%	100.0%	11.7%	10.7%	1.1%

The portfolio and the benchmark consist of five sectors and the allocation weights and returns are shown for each. Based on this data, without running the model, the portfolio over-performed the benchmark resulting in an active return or nominal alpha of 1.1%. The portfolio manager over-weighted Sector A relative to the benchmark sector and it appears to have paid off as the portfolio sector over-performed the benchmark sector return of 9.0% as well as the overall benchmark return of 10.7%. The portfolio manager under-weighted Sector C relative to the benchmark sector allocation, but Sector C's return was the highest performing sector for both the portfolio and the benchmark. Was under-weighting Sector C a good strategic decision? Did the manager take on additional risk to earn such a high return in Sector C? Sector E has the second highest allocation for both the portfolio and the benchmark, but the lowest

return for the portfolio and the second highest return for the benchmark. So, what does this all mean? As we will see, attribution analysis involves combining the portfolio and benchmark allocation sector weights and sector returns in an intuitive way to determine the effectiveness of the manager's active strategic allocation and selection decisions.

Exhibit III is not a required step, but it provides a good snapshot of the calculations that will be used to perform the Brinson-Fachler attribution analysis. You could skip this step and use the full complete formulas to calculate the components of allocation, selection, and interaction.

Exhibit III

A	B	H	I	J	K	L
		Diff in Portfolio and Benchmark Weights (C-D)	Diff in Portfolio and Benchmark Returns (E-F)	Diff in Weights * Diff in Returns (C-D)*(E-F)	Diff in Benchmark Sector Return vs. Overall Benchmark Return (F-\$F\$8)	Benchmark Weights (D)
10						
11	Sector A	30.0%	3.0%	0.9%	-1.7%	23.0%
12	Sector B	6.0%	-3.0%	-0.2%	-2.7%	2.0%
13	Sector C	-5.0%	3.0%	-0.2%	2.4%	14.0%
14	Sector D	-22.0%	-1.0%	0.2%	-0.6%	36.0%
15	Sector E	-9.0%	2.0%	-0.2%	1.4%	25.0%

The column headings are self-explanatory and will be used in the analysis shown in Exhibit IV.

Exhibit IV

A	B	M	N	O	P
17		BF Allocation (H*K)	BF Selection (I*L)	BF Interaction (J)	Nominal Alpha (M+N+O)
18	Sector A	-0.5%	0.7%	0.9%	1.1%
19	Sector B	-0.2%	-0.1%	-0.2%	-0.4%
20	Sector C	-0.1%	0.4%	-0.2%	0.2%
21	Sector D	0.1%	-0.4%	0.2%	0.0%
22	Sector E	-0.1%	0.5%	-0.2%	0.2%
23	Total	-0.8%	1.2%	0.6%	1.1%

Here we see that the active return or nominal alpha of 1.1% is due to a negative allocation score of -0.8%, a positive selection score of 1.2% and a positive interaction score of 0.6%. The -0.8% allocation score was primarily driven by Sector A where the manager over-weighted the sector relative to the benchmark sector. This was a poor strategic allocation decision because Sector A's 9.0% benchmark sector return as shown in Exhibit II, under-performed the overall benchmark of 10.7%. The portfolio return is not considered in allocation, but it is captured in selection. We clearly see that with the highest positive selection score of 0.7% in Sector A, the primary driver of the overall 1.2% selection score. The manager therefore made good individual investment selections in Sector A.

This analysis only tells us half the story; however, because it only incorporates weights and returns. It does not consider whether the manager took on risk more or less than the benchmark. Risk adjusted returns must be used in order to evaluate both returns and risk together. The analysis may be different depending on how risk is defined. For example, if you have a diversified portfolio, you could just focus on market risk,

otherwise known as systematic risk because non-systematic or idiosyncratic risk can be diversified away. If, however; your portfolio is not diversified, you should focus on total risk that combines both systematic and non-systematic risk. We will demonstrate this in two separate steps, first an analysis of systematic risk and then an analysis of total risk.

The price of risk as defined by CAPM is the benchmark excess return divided by the standard deviation of the benchmark excess return. This is the Sharpe Ratio of the benchmark. It is the price per unit of risk. In order to get total risk, the price per unit is multiplied by the amount of risk which according to CAPM is the covariance of the portfolio and benchmark excess returns divided by the standard deviation of the benchmark excess returns. The formula for the price of systematic risk looks like this;

Another way to look at this is the benchmark excess return multiplied by beta, which is the covariance of the portfolio and benchmark excess returns divided by the variance of the benchmark excess returns.

$$\text{Systematic Risk} = \text{Benchmark Excess Return} * \text{Beta}$$

Beta can also be calculated as the correlation multiplied by the standard deviation of the portfolio excess returns divided by the standard deviation of the benchmark excess returns.

This formula presentation will be used because it highlights the different risk adjusted attribution models we are proposing, one model that adjusts for systematic risk (beta) and another model that adjusts for a more comprehensive risk (total risk) that includes both systematic and non-systematic risk.

Exhibit V are therefore the data necessary to risk adjust the portfolio returns for both systematic or beta Risk and total risk.

It is important to note that the systematic risk for each sector must be measured relative to the overall benchmark return (weighted average of each sector) and not the benchmark return for each sector. Risk is measured relative to the “market” portfolio with the benchmark a proxy for the market. In this way, the overall beta is the weighted average of the beta for each sector as it should be. We believe Bacon errors by calculating a risk adjustment using the sector benchmark return with the result that the total risk adjustment does not reconcile with an overall beta.

Exhibit V

A	B	E	Q	R	S	T	U	F (cell \$F\$8)	V
25		Portfolio Returns	Correlation with the Overall Benchmark Return	Std Dev Portfolio Excess Returns	Std Dev Benchmark Excess Returns	Risk Free Rate	Beta (Q*R/S)	Overall Benchmark Return	Additional Market Risk Adjusted Return E-(\$F\$8- T)*(U-1)
26	Sector A	12.0%	0.65	9.0%	4.0%	1.0%	1.46	10.7%	7.5%
27	Sector B	5.0%	0.75	7.0%	2.0%	1.0%	2.63	10.7%	-10.7%
28	Sector C	16.0%	0.80	10.0%	5.0%	1.0%	1.60	10.7%	10.2%
29	Sector D	9.0%	0.50	6.0%	1.0%	1.0%	3.00	10.7%	-10.3%
30	Sector E	14.0%	0.90	5.0%	3.0%	1.0%	1.50	10.7%	9.2%
31	Total (Weighted average)	11.7%	0.69	7.9%	2.8%	1.0%	1.79		4.1%

Each sector's actual return is de-risked or risk adjusted for the amount of beta risk greater than or less than 1. The formula for this additional market risk is consistent with the calculation of beta risk discussed above for CAPM. The additional market risk is then subtracted from the actual return to get the risk adjusted return that is shown in column V.

1

The additional market risk adjusted returns calculated in column V can now be substituted in the Brinson model for the actual return that was shown in column E of

Exhibit II. The basic data necessary for calculating risk adjusted performance attribution is now shown in Exhibit VI.

Exhibit VI

A	B	C	D	V	F
				Additional Market Risk Adjusted Portfolio Returns	
33		Portfolio Weights	Benchmark Weights		Benchmark Returns
34	Sector A	53.0%	23.0%	7.5%	9.0%
35	Sector B	8.0%	2.0%	-10.7%	8.0%
36	Sector C	9.0%	14.0%	10.2%	13.0%
37	Sector D	14.0%	36.0%	-10.3%	10.0%
38	Sector E	16.0%	25.0%	9.2%	12.0%
39	Total	100.0%	100.0%	4.1%	10.7%

The risk adjusted attribution scores are therefore shown in Exhibit VII;

Exhibit VII

A	B	M	W	X	Y
			Risk Adjusted BF Selection	Risk Adjusted BF Interaction	
41		BF Allocation			<i>Jensen's Alpha</i>
42	Sector A	-0.5%	-0.3%	-0.4%	-1.3%

43	Sector B	-0.2%	-0.4%	-1.1%	-1.7%
44	Sector C	-0.1%	-0.4%	0.1%	-0.4%
45	Sector D	0.1%	-7.3%	4.5%	-2.7%
46	Sector E	-0.1%	-0.7%	0.3%	-0.6%
47	Total	-0.8%	-9.1%	3.3%	-6.6%

And can now be compared to the nominal attribution scores previously calculated in Exhibit IV as shown in Exhibit VIII.

Exhibit VIII

A	B	P	Y	Z
49		Nominal Alpha	Jensen's Alpha	Additional Market Risk
50	Sector A	1.1%	-1.3%	2.4%
51	Sector B	-0.4%	-1.7%	1.3%
52	Sector C	0.2%	-0.4%	0.5%
53	Sector D	0.0%	-2.7%	2.7%
54	Sector E	0.2%	-0.6%	0.8%
55	Total	1.1%	-6.6%	7.6%

On a nominal basis, the portfolio over-performed the benchmark by 1.1%, primarily driven by Sector A. On an additional market risk adjusted basis the portfolio under-performed the benchmark by -6.6%, primarily driven by Sector's D, B and A's poor risk-adjusted performance. The additional market risk is the difference between nominal alpha and the risk-adjusted Jensen's alpha. This highlights the additional risk of picking investments that are risky than the benchmark.

The approach discussed above to risk adjust returns is done for each sector for the attribution analysis. This can be done using either the BF or BHB versions. We prefer BF for the following reasons: First, BF provides more insight into the interpretation of the allocation component for individual sectors without impacting the selection component. For example, it will be positive if a sector is overweighed that has a benchmark return that is higher than the average of all sector returns. The BHB model only requires the sector benchmark return be positive to have a positive score for the sector. Our risk adjusted attribution approach complements BF nicely by providing a better interpretation of the selectivity score for each sector without impacting the allocation scores. For example, the selection component might be positive for a sector before risk adjusting but lower or negative after risk adjusting the sector.

III. Adjusting Alpha for Diversification

Fama (1972) shows how to further decompose Jensen's alpha into the portion that is due to having less diversification than the benchmark. As he pointed out, this is especially relevant if the portfolio is the only one held by the investor and thus the only source of diversification. The idea is that in addition to being compensated for beta (systematic) risk, the investor needs compensation for taking on more unsystematic risk by deviating from the benchmark which is presumably well diversified. The required return for taking on this additional risk is calculated as follows:

$$\text{Risk Premium for non-Diversification} = (\beta_F - \beta_P) (R_B - R_F).$$

Where β_F has been referred to as "Fama's beta." Fama's beta can be calculated by taking the regular beta and dividing by the correlation coefficient. Alternatively, it is simply the ratio of the standard deviation of the portfolio to the standard deviation of the benchmark.

The expected return with this additional risk adjustment is therefore as follows:

$$\text{Expected Return} = R_F + (R_B - R_F) \beta_P + (\beta_F - \beta_P) (R_B - R_F)$$

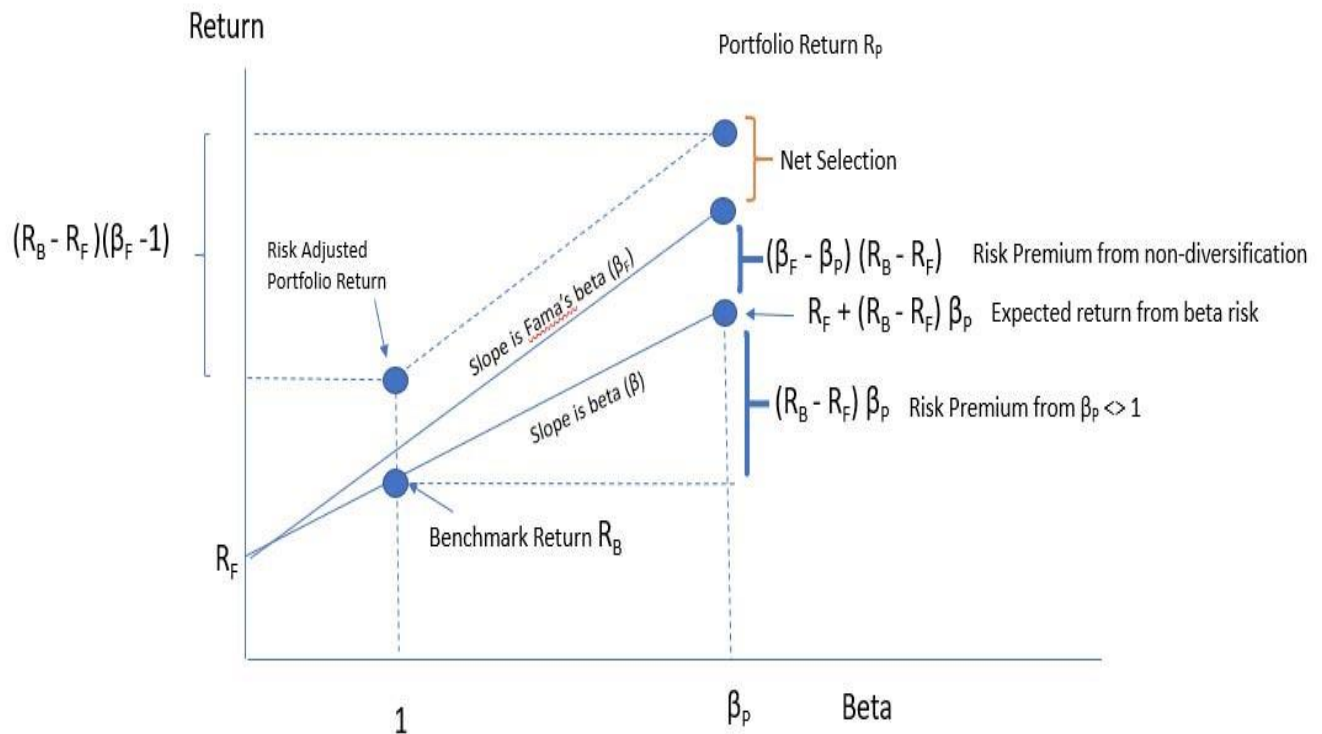
The difference between the actual portfolio returns and this expected return is referred to as “net selectivity” which includes a more comprehensive risk adjustment that might be relevant for many investors. To get the more comprehensive risk adjusted return, we now have Fama Risk-adjusted return = $(R_B - R_F) (\beta_F - 1)$ where Fama’s beta is used in place of the regular beta.

The more comprehensive portfolio risk adjusted return can now be substituted for the nominal portfolio return in the Brinson attribution analysis in the same way that was discussed previously with the beta risk adjusted return.

Exhibit IX illustrates the incorporation of this additional risk adjustment in the model. The slope is steeper with Fama’s beta as the measure of risk. (Recall that it is regular beta divided by the correlation coefficient, so it must have a greater slope if the correlation is less than 1.) The resulting risk adjusted return will be lower than it would have been by lowering the return in parallel with the regular beta that has a smaller slope as was illustrated in Exhibit I.

Exhibit IX

Risk Adjusted Portfolio Return



As noted above, the difference between the regular beta that reflects systematic risk and the use of Fama beta that includes the impact of having less diversification depends on the correlation between the portfolio and the benchmark. Fama beta is the regular beta divided by the correlation coefficient. If the two were perfectly correlated, there would be no difference because this would mean there is no difference in diversification. The lower the correlation, the greater the difference because the portfolio is lesser and lesser diversified.

Exhibit X shows how to calculate Fama beta which is the standard deviation of the portfolio excess returns divided by the standard deviation of the benchmark excess returns. This is the total risk of not achieving the benchmark returns.

Exhibit X

A	B	E	R	S	T	AA	F (cell \$F\$8)	AC
25		Portfolio Returns	Std Dev Portfolio Excess Returns	Std Dev Benchmark Excess Returns	Risk Free Rate	Fama Beta (R/S)	Overall Benchmark Return	Fama Adjusted Return E-(\$F\$8- T)*(AA-1)
26	Sector A	12.0%	9.0%	4.0%	1.0%	2.25	10.7%	-0.1%
27	Sector B	5.0%	7.0%	2.0%	1.0%	3.50	10.7%	-19.1%
28	Sector C	16.0%	10.0%	5.0%	1.0%	2.00	10.7%	6.4%
29	Sector D	9.0%	6.0%	1.0%	1.0%	6.00	10.7%	-39.3%
30	Sector E	14.0%	5.0%	3.0%	1.0%	1.67	10.7%	7.6%
31	Total (weighted average)	11.7%	7.9%	2.8%	1.0%	2.76		-5.3%

Just as we substituted the beta risk adjusted returns for the actual returns, we can substitute the Fama beta risk adjusted returns for the actual returns. The following Exhibit XI shows the result of that. **Exhibit XI**

A	B	C	D	AB	F
33		Portfolio Weights	Benchmark Weights	Fama Beta Risk Adjusted Portfolio Returns	Benchmark Returns
34	Sector A	53.0%	23.0%	-0.1%	9.0%
35	Sector B	8.0%	2.0%	-19.1%	8.0%
36	Sector C	9.0%	14.0%	6.4%	13.0%
37	Sector D	14.0%	36.0%	-39.3%	10.0%
38	Sector E	16.0%	25.0%	7.6%	12.0%
39	Total	100.0%	100.0%	-5.3%	10.7%

The total risk adjusted results for attribution are now depicted in Exhibit XII as follows;

Exhibit XII

A	B	W	AC	AD	AE
41		BF Allocation	Risk Adjusted BF Selection	Risk Adjusted BF Interaction	Fama's Alpha
42	Sector A	-0.5%	-2.1%	-2.7%	-5.3%
43	Sector B	-0.2%	-0.5%	-1.6%	-2.3%
44	Sector C	-0.1%	-0.9%	0.3%	-0.7%
45	Sector D	0.1%	-17.7%	10.8%	-6.8%
46	Sector E	-0.1%	-1.1%	0.4%	-0.8%
47	Total	-0.8%	-22.4%	7.2%	-15.9%

Just as we were able to determine the additional market risk adjustment by comparing nominal alpha to Jensen's alpha, we can also determine the Fama beta risk adjustment by comparing nominal alpha to Fama alpha and then adjusting for the additional market risk identified using Jensen's alpha as shown in Exhibit XIII.

Exhibit XIII

A	B	C	AE	AF	Z	AH
49		Nominal Alpha	Fama's Alpha	Total Risk (C- AE)	Additional Market Risk	Non- Diversification Risk (AF- Z)
50	Sector A	1.1%	-5.3%	6.4%	2.4%	4.0%
51	Sector B	-0.4%	-2.3%	1.9%	1.3%	0.7%
52	Sector C	0.2%	-0.7%	0.9%	0.5%	0.3%
53	Sector D	0.0%	-6.8%	6.8%	2.7%	4.1%
54	Sector E	0.2%	-0.8%	1.0%	0.8%	0.3%
55	Total	1.1%	-15.9%	17.0%	7.6%	9.4%

As proof of this model, we can recalculate both the additional market risk as well as the non-diversification risk premium (non-systematic risk) as shown in Exhibit XIV.

Exhibit XIV

Calculation of Additional Beta Risk and Non-Diversification Risk Premiums	Benchmark
Benchmark Return	10.65%
Risk Free Rate	1.00%
Benchmark Excess Return	9.65%
Beta - 1	0.79
Additional Beta Risk Premium	7.62%
Fama Beta	2.76
Beta	1.79
Fama Beta - Beta	0.97
Benchmark Excess Returns	9.65%
Non-Diversification Risk Premium	9.36%
Total Systematic and Non-Systematic Risk	★ 16.98%

The first step to calculate the additional beta risk is to multiply the benchmark excess return by the portfolio beta greater than or less than 1. In this case the $9.65\% \times .79$ or 7.62%. This is the additional market risk premium which is the risk of selecting risky investments than the benchmark. The second step is to calculate the risk of non-diversification using Fama's beta which is the difference between Fama's beta and beta multiplied by the benchmark excess return. In this case the $.97 \times 9.65\%$ or 9.36%. This is the risk premium required for not diversifying like the benchmark. The sum of the market risk of 7.62% and the non-market risk of 9.36% is a total comprehensive risk of 16.98%. See Appendix II for a summary presentation of the results of the Risk Adjusted Attribution Analysis model we have proposed.

The following Exhibit XV shows the overall attribution analysis on a nominal basis, additional market risk adjusted basis and Fama risk adjusted basis.

Exhibit XV

Risk	Adjusted				
Attribution Analysis	Total	Allocation	Selection	Interaction	
Nominal Alpha	1.1%	-0.8%	1.2%	0.6%	
Additional Market Risk	-7.6%	0.0%	-10.3%	2.7%	
Jensen's Alpha	-6.6%	-0.8%	-9.1%	3.3%	
Non-Diversification Risk	-9.4%	0.0%	-13.3%	3.9%	
Fama Alpha	-15.9%	-0.8%	-22.4%	7.2%	

The portfolio manager made some poor strategic allocation decisions. You may recall that the portfolio manager made a large bet in Sector A with an allocation of more than 50%. The negative allocation score of -0.8% is the original nominal allocation because it is not risk-adjusted based on the formulas in the Brinson model. The overall negative -22.4% selection indicates that the manager picked some bad investments. On a nominal basis the selection score was a positive 1.2%, which gives a false reading. After adjusting for the additional market risk of a beta greater than or less than 1, the selection score deteriorated significantly to a negative -10.3%. This indicates the manager picked riskier investments than the benchmark. We see this is Sector's D, B and A with negative selection scores of -2.7%, -1.7% and -1.3% as shown in column Z "Jensen's alpha" of Exhibit VII. After considering total risk using Fama's beta we see a further decline in selection of negative -13.3%. This is since the manager made big bets in a particular sector and therefore sacrificed diversification. The non-diversification risk premium of -9.4% reflects this poor decision.

The details by sector can be presented as shown in Exhibit XVI.

Exhibit XVI

Risk	Adjusted	Sector A	Sector B	Sector C	Sector D	Sector E	Total
Attribution Analysis							
Nominal Alpha Allocation		-0.5%	-0.2%	-0.1%	0.1%	-0.1%	-0.8%

Total Allocation	-0.5%	-0.2%	-0.1%	0.1%	-0.1%	-0.8%
Nominal Alpha Selection	0.7%	-0.1%	0.4%	-0.4%	0.5%	1.2%
Beta Risk Selection	-1.0%	-0.3%	-0.8%	-6.9%	-1.2%	-10.3%
Fama Beta Selection	-1.7%	-0.2%	-0.5%	-10.4%	-0.4%	-13.3%
Total Selection	-2.1%	-0.5%	-0.9%	-17.7%	-1.1%	-22.4%
Nominal Interaction	0.9%	-0.2%	-0.2%	0.2%	-0.2%	0.6%
Beta Risk Interaction	-1.3%	-0.9%	0.3%	4.2%	0.4%	2.7%
Fama Beta Risk Interaction	-2.3%	-0.5%	0.2%	6.4%	0.1%	3.9%
Total Interaction	-2.7%	-1.6%	0.3%	10.8%	0.4%	7.2%
Fama Alpha	-5.3%	-2.3%	-0.7%	-6.8%	-0.8%	-15.9%
Nominal Alpha	1.1%	-0.4%	0.2%	0.0%	0.2%	1.1%
Jensen's Alpha	-1.3%	-1.7%	-0.4%	-2.7%	-0.6%	-6.6%
Additional Market Risk Premium (Nominal Alpha - Jensen's Alpha)	2.4%	1.3%	0.5%	2.7%	0.8%	7.6%
Non-Diversification Risk Premium (Fama Alpha - Jensen's Alpha)	4.0%	0.7%	0.3%	4.1%	0.3%	9.4%
Total Risk Premium	6.4%	1.9%	0.9%	6.8%	1.0%	17.0%
Nominal Alpha - Total Risk Premium = Fama's Alpha	-5.3%	-2.3%	-0.7%	-6.8%	-0.8%	-15.9%

This detail allows the reader to see the nominal attribution, the incremental effects of both additional market risk as well as the non-diversification risk premiums by sector by management effects of allocation, selection, and interaction.

III. MM vs. Fama beta: Both MM and Fama beta are ways of getting a risk adjusted return. One might wonder if they result in the same risk adjusted return. The difference between using MM vs Fama beta to get a risk adjusted return can be calculated as follows:

$$\text{MM} - \text{Fama} = (R_P - R_F) (\sigma_B / \sigma_P) + (R_B - R_F) (\sigma_P / \sigma_B)$$

Note that the terms will be opposite in sign with one being negative depending on which standard deviation is larger (portfolio or benchmark). The difference will be zero if the two standard deviations are the same. The greater the difference in standard deviations, the greater the difference in risk adjusted returns. The use of the Fama beta to calculate the risk adjusted return is the only one that allows us to separate the non-diversification impact by comparing with the expected return using the traditional beta.

IV. A Note on Interaction

In addition to allocation and selection components, attribution analysis has an interaction term that is the difference in weights times the difference in returns. Some analysts prefer to combine this term with either allocation or selection rather than showing it separately. Rather than take a position on this debate, we simply want to point out that if it is kept as a separate term, it does have some interpretation in our risk adjusted attribution analysis. The interaction term will be more positive (or less negative) if the manager has either over-weighted a less risky sector or under-weighted a riskier sector. Vice versa, the interaction term will be less positive or more negative if the manager has over-weighted a riskier sector or under-weighted a less risky sector.

V. Conclusion

We have developed a risk adjusted performance attribution analysis that integrates risk measures with the Brinson models of attribution which allows us to decompose the excess portfolio return into components of risk, allocation and net selection that is additive and consistent with financial theory. Either the beta traditionally used to calculate Jensen's alpha or the beta proposed by Fama to incorporate unsystematic risk can be used to risk adjust the returns that are incorporated into the Brinson attribution analysis. The result is that the difference between the portfolio and benchmark return is decomposed into the following components:

1. Risk free rate
2. Risk premium due to the portfolio beta
3. Risk premium due to lack of diversification (optional)
4. Net selection
5. Allocation
6. Interaction

The third component can be ignored if it is believed that the investor has other sources of diversification so deviating from the benchmark diversification isn't relevant. In this case, that portion of the risk premium becomes part of the selection component of attribution analysis in step 4. Allocation in step 5 is not impacted by risk adjustments because the benchmark return is used to determine allocation impacts in attribution analysis. And the risk adjustments are for additional market risk above (or below) the benchmark. Note, however that the interaction term does include the impacts of risk adjustment and will be greater (lower) if the portfolio had a greater (lesser) allocation to a riskier sector and vice versa.

Risk adjusted performance attribution can give us quite different interpretation of which sectors contributed to better or worse performance relative to the benchmark. Traditional attribution analysis could result in a manager appearing

to have done well in a sector where the higher return relative to the benchmark was just due to taking on more risk whereas other sectors may be where the manager was able to outperform the benchmark. Or the manager could appear to have underperformed in a sector that was a less risky sector.

This article has used single period examples to illustrate the concept. But it can easily be extended to a multi-period analysis as is done with traditional attribution analysis.

Although our focus has been on a risk adjustment model that is consistent with the CAPM, it should be noted that beta is simply the sensitivity of a portfolio's return to the benchmark's return. It is the slope of a simple linear regression that explains the portfolio return (or excess return over the risk-free rate) by the benchmark return (or excess return). Thus, even if one questions the validity of the CAPM assumptions, risk adjusting by beta and doing attribution analysis of Jensen's alpha is still relevant. Furthermore, the use of Fama's beta is only focused on total risk which is just uses the standard deviations of the portfolio and the benchmark. The point is that risk adjustment needs to be formally integrated into attribution analysis as we have done.

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APPENDIX I Math Proof

1. Calculate Risk Adjusted Portfolio Return

R_{pA} is the Risk Adjusted portfolio (or fund) return

$$R_{pA} = R_B + \alpha$$

$$\alpha = R_p - \text{Expected } R_p$$

$$\text{Expected } R_p = R_F + (B - R_F) \times \beta_p$$

$$R_{pA} = R_B + R_p - (R_F + (B - R_F) \times \beta_p)$$

$$R_{pA} = R_p + (B - R_F) - (B - R_F) \times \beta_p$$

$$R_{pA} = (B - R_F)(1 - \beta_p) + R_p$$

$$R_{pA} = R_p - (B - R_F)(\beta_p - 1)$$

R_{pA} – Risk adjusted portfolio return for sector

R_B – Benchmark return for sector

R_p – Portfolio return for sector

R_F – Risk free rate

β_p – Beta for the sector relative to overall benchmark

W_p – Portfolio weight for sector

W_B – Benchmark weight for sector

B – Overall Benchmark weight (wtd. ave. of sectors)

α – Jensen's alpha

2. Brinson-Fachler attribution analysis using Risk Adjusted Portfolio Returns

$$R_{pA} - R_B = \sum (R_{pA} - R_B) W_B + \sum (W_p - W_B) (R_B - B) + \text{Interaction}$$

Substitute $R_{pA} = R_p - (B - R_F)(\beta_p - 1)$ in above equation

$$R_{pA} - R_B = \sum (R_p - (B - R_F)(\beta_p - 1) - R_B) W_B + \sum (W_p - W_B) (R_B - B) + \text{Interaction}$$

$$R_{pA} - R_B = \sum (R_p - (B - R_F) \times \beta_p + R_B - R_F - R_B) W_B + \sum (W_p - W_B) (R_B - B) + \text{Interaction}$$

$$R_{pA} - R_B = \sum (R_p - R_F - (B - R_F) \beta_p) W_B + \sum (W_p - W_B) (R_B - B) + \text{Interaction}$$

$$\text{Or } R_{pA} - R_B = \sum (R_p - [R_F + (B - R_F) \beta_p]) W_B + \sum (W_p - W_B) (R_B - B) + \text{Interaction}$$

Jensen's alpha for each sector

APPENDIX II Sample Summary Report

Risk Analysis	Actual Return	Benchmark	Difference	Comments
Actual Portfolio Return vs. Benchmark	11.70%	10.65%	1.05%	Nominal Alpha
Less Additional Market Risk Premium	-7.62%	0.00%	-7.62%	Market (Systematic) Risk
Portfolio Beta Risk Adjusted Return	4.08%	10.65%	-6.57%	Jensen's Alpha
Less Non-Diversification Risk Premium	-9.36%	0.00%	-9.36%	Non-Diversification (Unsystematic) Risk
Portfolio Total Risk Adjusted Return	-5.28%	10.65%	-15.93%	Fama Beta Alpha
Adjust for Risk	16.98%	7.62%	9.36%	
Actual Portfolio Return vs. Expected Return	11.70%	18.27%	-6.57%	Reconciliation to CAPM Jensen's Alpha
Risk Adjusted Attribution Analysis	Total	Allocation	Selection	Interaction
Nominal Alpha	1.1%	-0.8%	1.2%	0.6%
Additional Market Risk	-7.6%	0.0%	-10.3%	2.7%
Jensen's Alpha	-6.6%	-0.8%	-9.1%	3.3%
Non-Diversification Risk	-9.4%	0.0%	-13.3%	3.9%
Fama Alpha	-15.9%	-0.8%	-22.4%	7.2%

Calculation of Additional Beta Risk and

Benchmark Return	10.65%
Risk Free Rate	1.00%
Benchmark Excess Return	9.65%
Beta - 1	0.79

Additional Market Risk Premium	7.62%
Fama Beta	2.76
Beta	1.79
Fama Beta - Beta	0.97
Benchmark Excess Returns	9.65%
Non-Diversification Risk Premium	9.36%
Total Systematic and Non- Systematic Risk	16.98%